

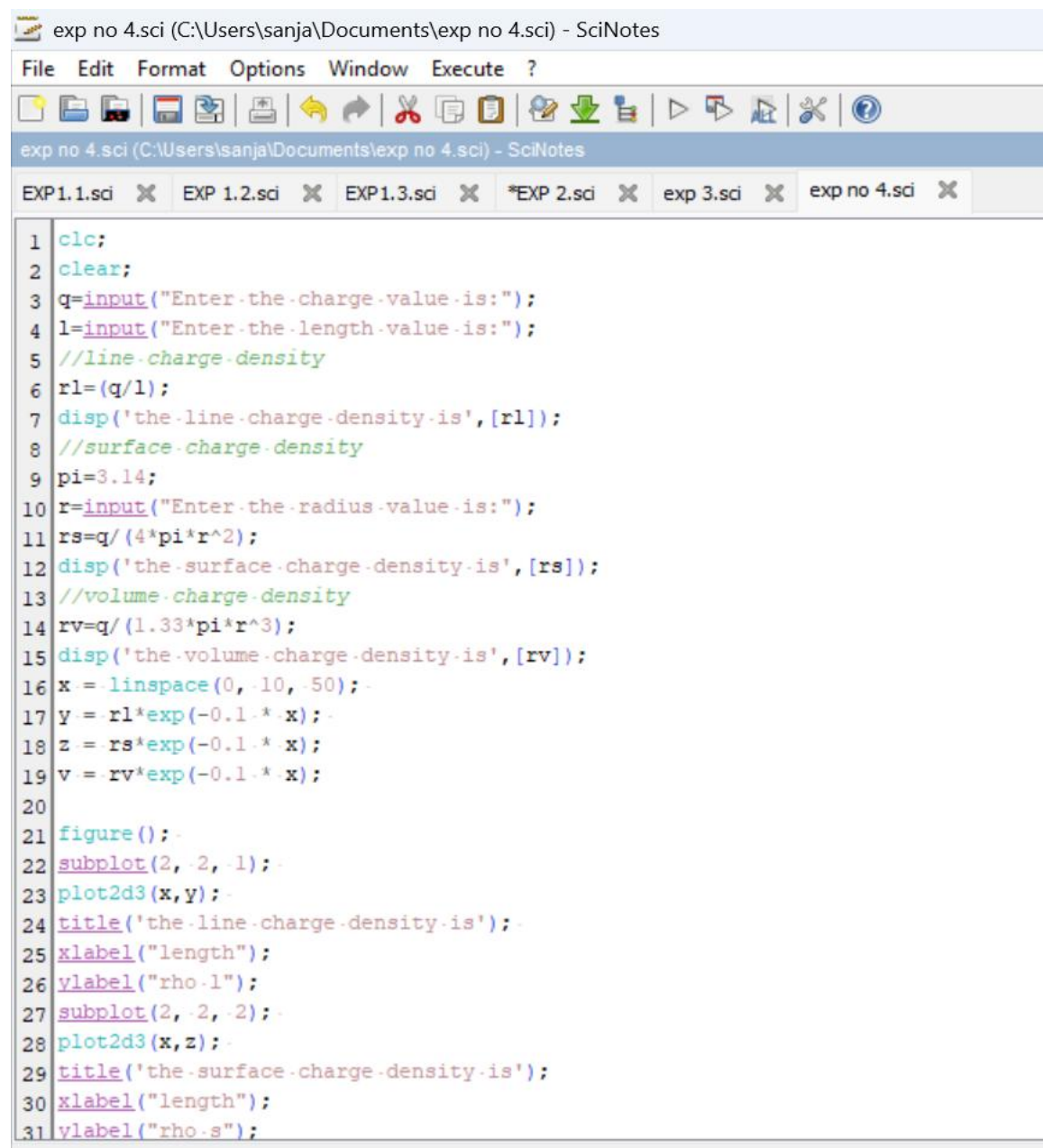
**Exp. No: 4** Analysis of Continuous Charge Distribution in electric field. To find the (i) line charge density, (ii) Surface charge density and (iii) Volume charge density for any specific (a) line, (b) surface or (c) volume.

**Aim:** To write program for continuous Charge Distribution in electric field. The different types of charge density there are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density using SCILAB.

### Software required:

- SCILAB version 6.0.1

### Partical calucation



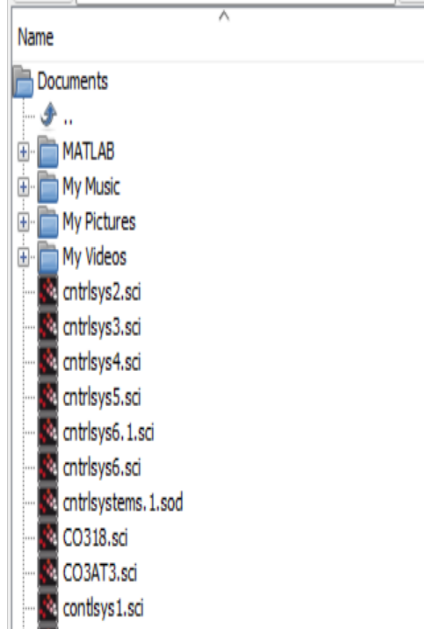
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exp no 4.sci (C:\Users\sanja\Documents\exp no 4.sci) - SciNotes
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exp no 4.sci (C:\Users\sanja\Documents\exp no 4.sci) - SciNotes
EXP1.1.sci X EXP 1.2.sci X EXP1.3.sci X *EXP 2.sci X exp 3.sci X exp no 4.sci X
1 clc;
2 clear;
3 q=input("Enter the charge value is:");
4 l=input("Enter the length value is:");
5 //line charge density
6 rl=(q/l);
7 disp('the line charge density is',[rl]);
8 //surface charge density
9 pi=3.14;
10 r=input("Enter the radius value is:");
11 rs=q/(4*pi*r^2);
12 disp('the surface charge density is',[rs]);
13 //volume charge density
14 rv=q/(1.33*pi*r^3);
15 disp('the volume charge density is',[rv]);
16 x=-linspace(0,-10,-50);
17 y=-rl*exp(-0.1.*x);
18 z=-rs*exp(-0.1.*x);
19 v=-rv*exp(-0.1.*x);
20
21 figure();
22 subplot(2,-2,-1);
23 plot2d3(x,y);
24 title('the line charge density is');
25 xlabel("length");
26 ylabel("rho.l");
27 subplot(2,-2,-2);
28 plot2d3(x,z);
29 title('the surface charge density is');
30 xlabel("length");
31 ylabel("rho.s");
```

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File Browser

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Scilab 2025.1.0 Console

Enter the charge value is:  $5 \times 10^{-6}$

Enter the length value is: 5

"the line charge density is"

0.000001

Enter the radius value is: 2

"the surface charge density is"

9.952D-08

"the volume charge density is"

0.0000001

exec: Wrong number of output argument(s): 0 expected.

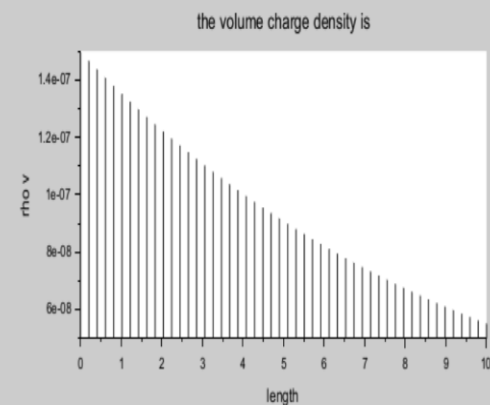
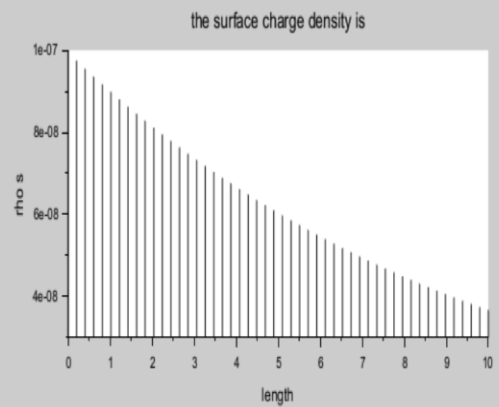
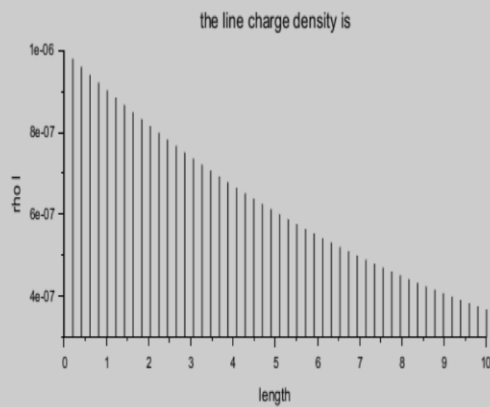
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### Theoretical calculation:

Handwritten calculations on a piece of paper. At the top, 'o/p v' is written in red with a checkmark. The calculations are as follows:

$$\begin{aligned} 4) \text{ charge } (q) &= 5 \times 10^{-6} \\ \text{length } (L) &= 5 \text{ cm} \\ \text{radius } (r) &= 2 \text{ mm} \end{aligned}$$
$$r_1 = (q/L) = \frac{5 \times 10^{-6}}{5} = 1 \times 10^{-6}$$
$$r_s = \frac{q}{4\pi r^2} = \frac{(5 \times 10^{-6})}{4\pi (2 \times 10^{-3})^2} = 9.95 \times 10^{-8}$$
$$r_v = \frac{q}{\frac{4}{3}\pi r^3} = \frac{5 \times 10^{-6}}{\frac{4}{3}\pi (2 \times 10^{-3})^3} = 1.4 \times 10^{-7} \quad 149.657$$

At the bottom, 'o/p v' is written in red with a checkmark.

### Result:

Thus the program was verified for continuous Charge Distribution in electric field. The different types of charge density there are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density using SCILAB was successfully